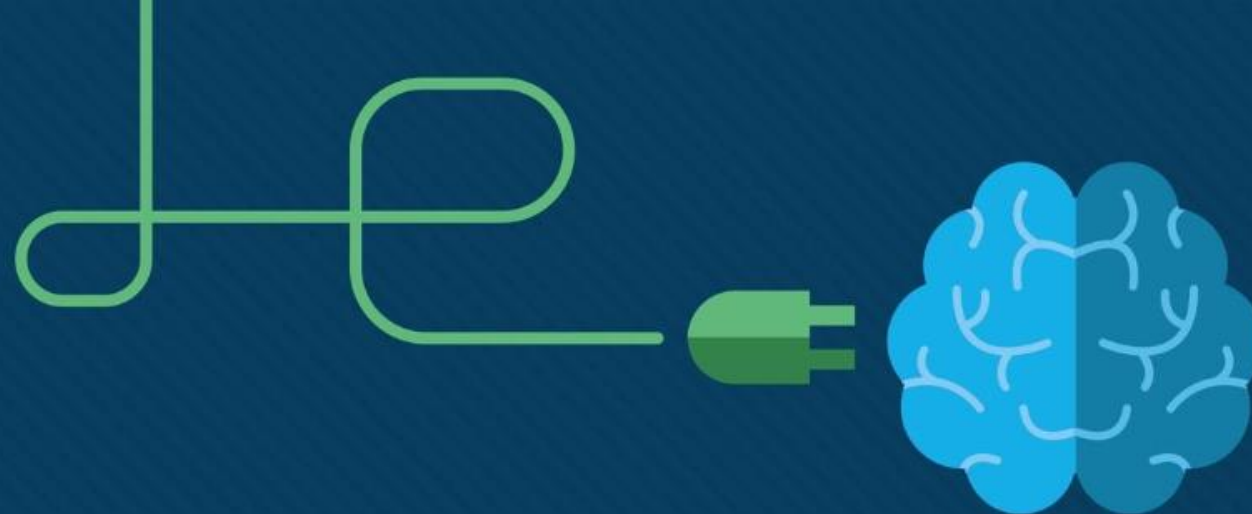




Chapter 1: Introduction to Networks



Chapter 1

- 1.1 Network Components and Topologies
- 1.2 Types of Networks
- 1.3 Internet Connections
- 1.4 Reliable networks
- 1.5 Network Trends
- 1.6 Issues of Network Management

Networks Affect Our Lives / Globally Connected

No Boundaries

- Advancements in networking technologies are helping create a **world without boundaries**.
- The immediate nature of communications over the Internet encourages **global communities**.
- Cisco refers to the impact of the Internet and networks on people the “**human network**”.



1.1 Network Components and Topologies

1.1.1 Network Components

Network Components

Host Roles

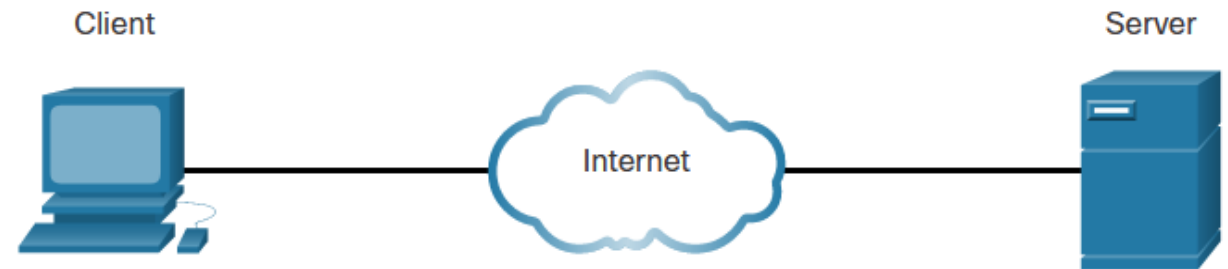
Every computer on a network is called a **host** or **end device**.

Servers are computers that **provide information** to end devices:

- email servers
- web servers
- file server

Clients are computers that **send requests** to the servers to retrieve information:

- web page from a web server
- email from an email server

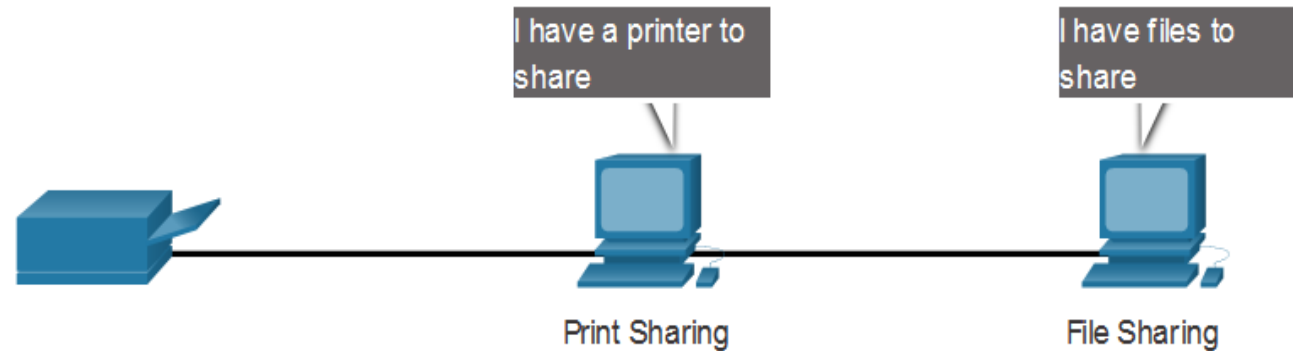


Server Type	Description
Email	Email server runs email server software. Clients use client software to access email.
Web	Web server runs web server software. Clients use browser software to access web pages.
File	File server stores corporate and user files. The client devices access these files.

Network Components

Peer-to-Peer

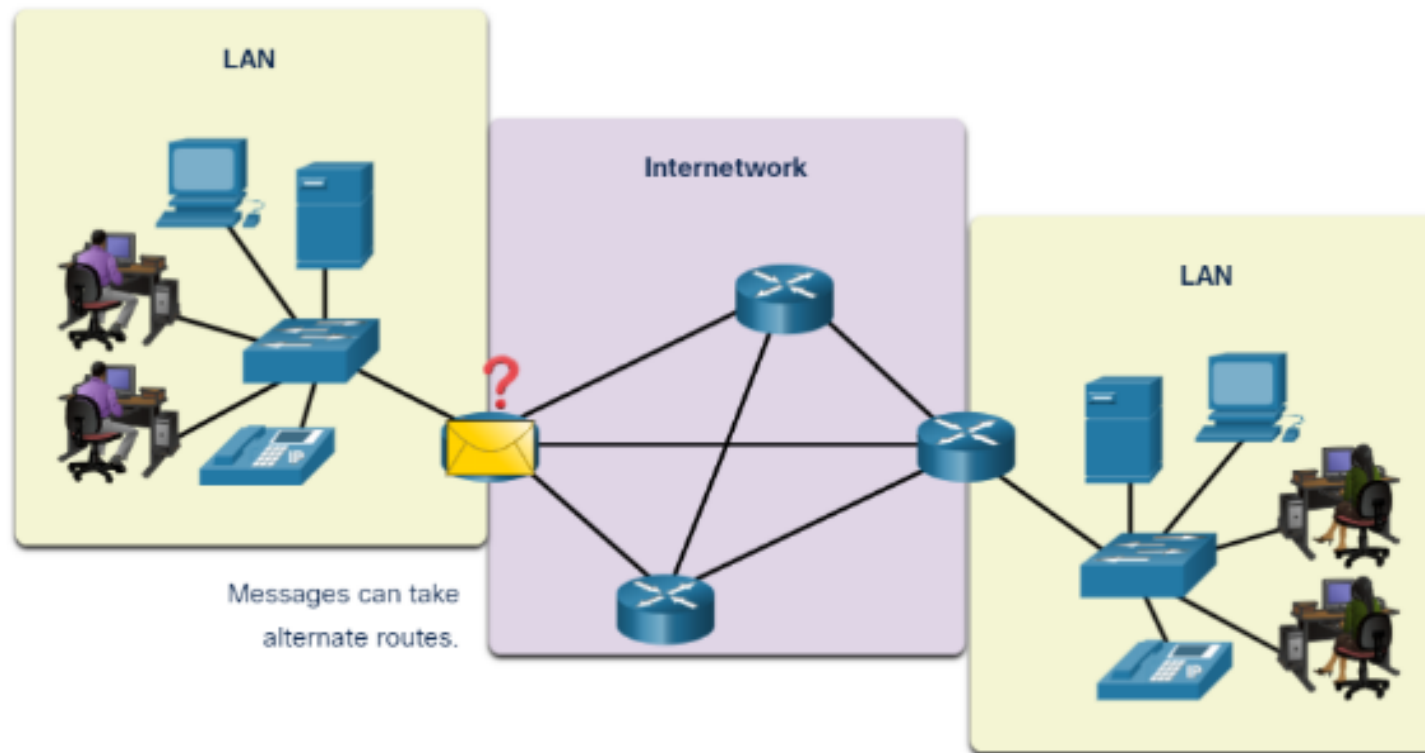
It is possible to have **a device be a client and a server** in a Peer-to-Peer Network. This type of network design is only recommended for **very small** networks.



Advantages	Disadvantages
Easy to set up	No centralized administration
Less complex	Not as secure
Lower cost	Not scalable
Used for simple tasks: transferring files and sharing printers	All devices may act as both clients & servers which can slow their performance

End Devices

An end device is where a **message originates from or where it is received**. Data originates with an end device, flows through the network, and arrives at an end device.

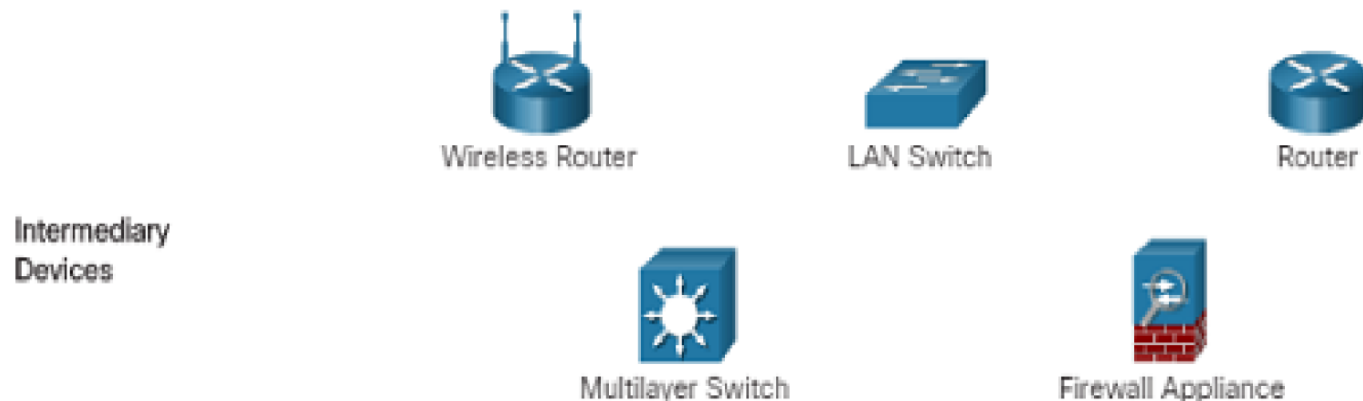


Intermediary Network Devices

An intermediary device **interconnects end devices**. Examples include switches, wireless access points, routers, and firewalls.

Management of data as it flows through a network is also the role of an intermediary device, including:

- Regenerate and retransmit data signals.
- Maintain information about what pathways exist in the network.
- Notify other devices of errors and communication failures.

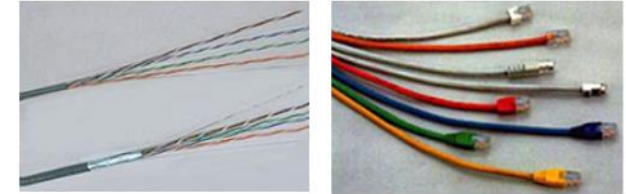


Network Media

Communication across a network is carried through a medium which **allows a message to travel from source to destination.**

Media Types	Description
Metal wires within cables	Uses electrical impulses
Glass or plastic fibers within cables (fiber-optic cable)	Uses pulses of light.
Wireless transmission	Uses modulation of specific frequencies of electromagnetic waves.

Copper



Fiber-optic



Wireless



1.1 Network Components and Topologies

1.1.2 Network Representations and Topologies

Network Representations and Topologies

Network Representations

Network diagrams, often called **topology diagrams**, use symbols to represent devices within the network.

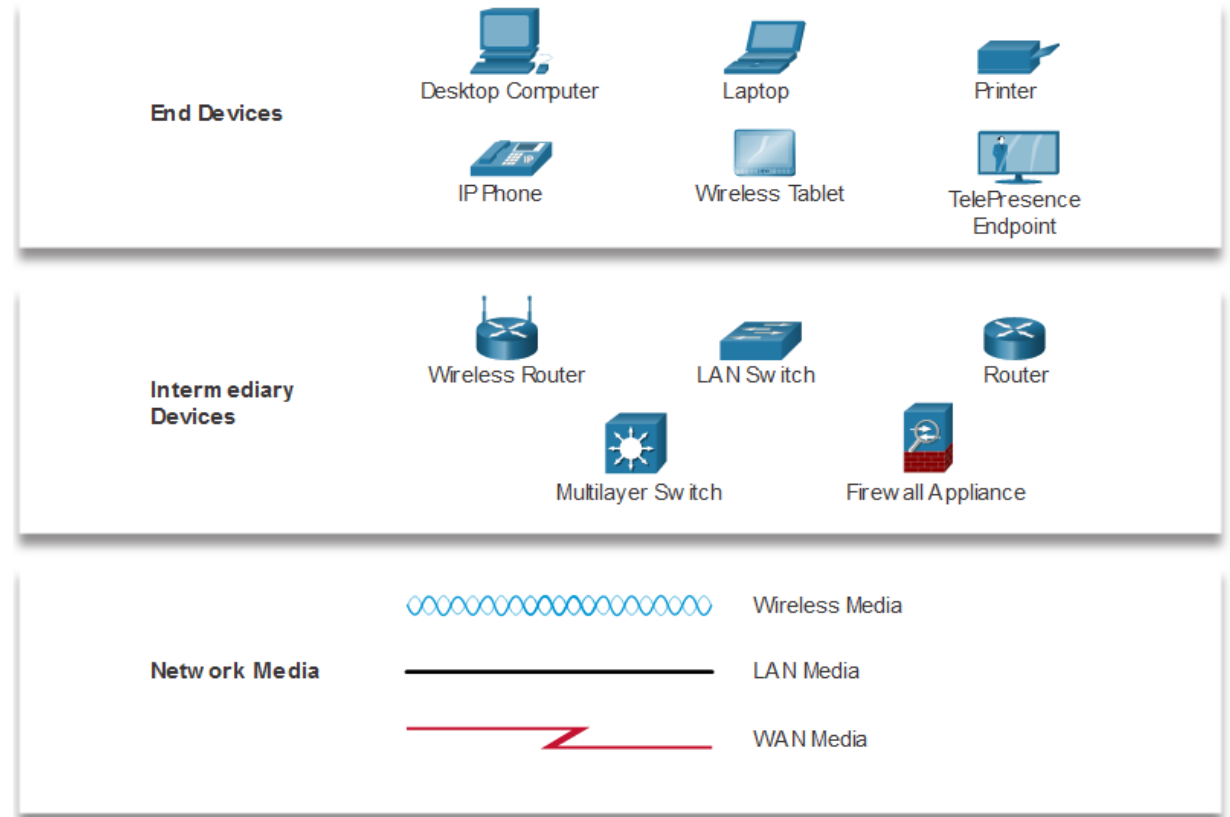
Important terms to know include:

Network Interface Card (NIC)

Physical Port

Interface

Note: Often, the terms port and interface are used interchangeably



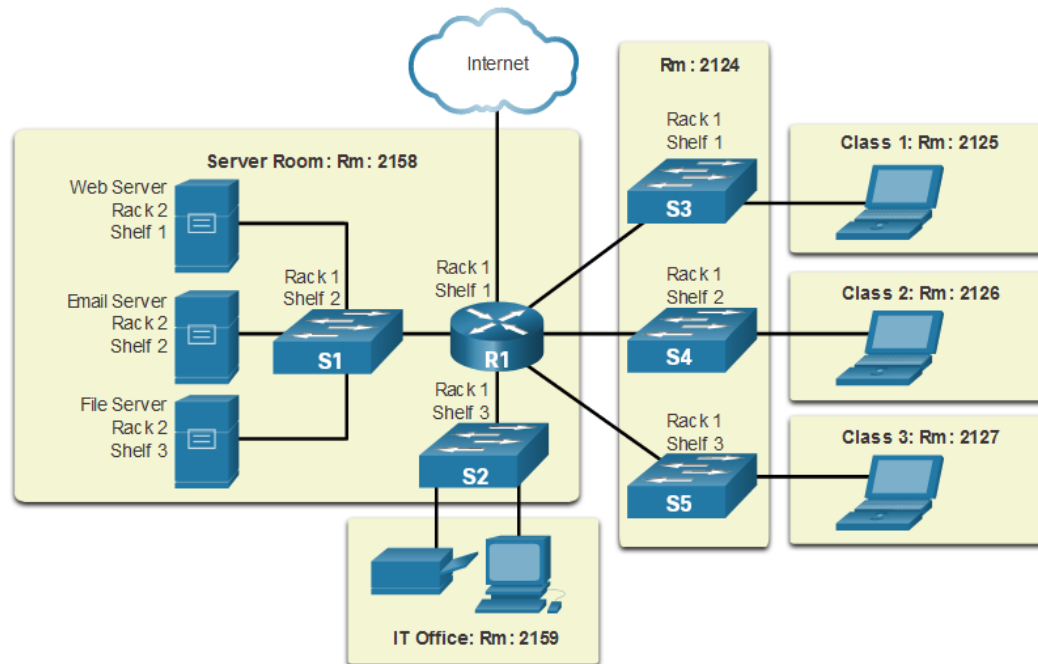
Network Representations

- **Network Interface Card** - A NIC, or LAN adapter, provides the physical connection to the network at the PC or other end device. The media connecting the PC to the networking device plugs directly into the NIC.
- **Physical Port** - A connector or outlet on a networking device where the media is connected to a host or other networking device.
- **Interface** - Specialized ports on an internetworking device that connect to individual networks. Because routers are used to interconnect networks, **the ports on a router are referred to network interfaces.**
- **Note:** Often, the terms port and interface are used interchangeably.

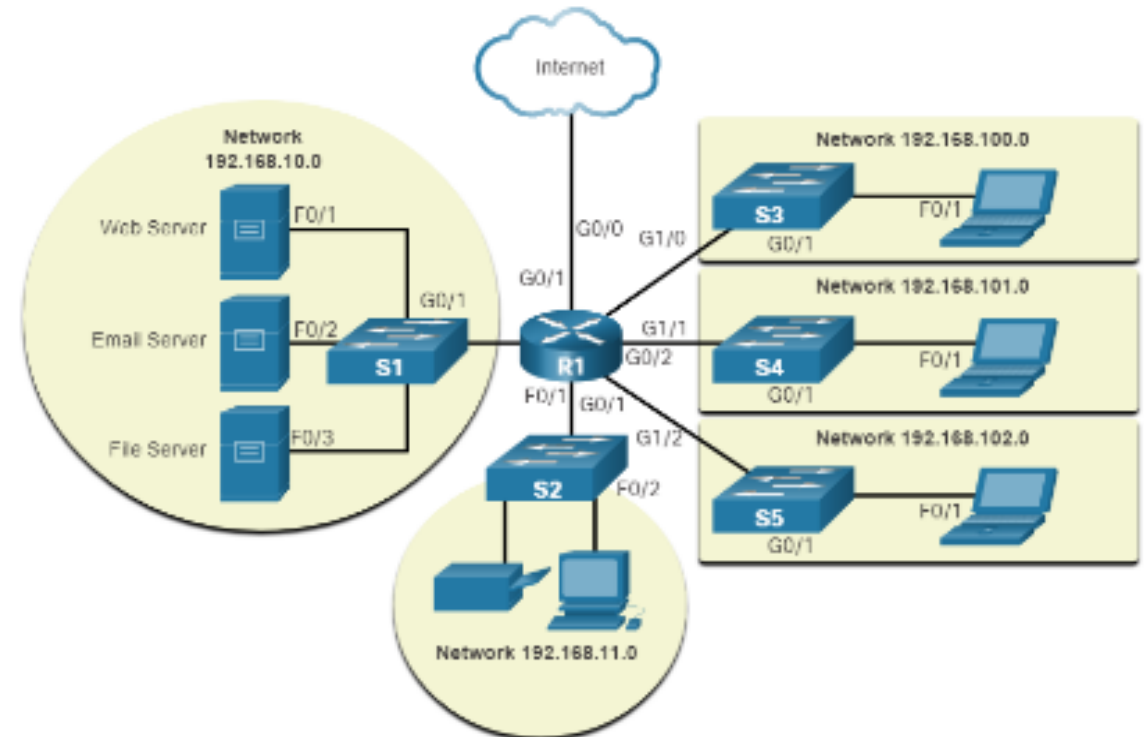
Network Representations and Topologies

Topology Diagrams

Physical topology diagrams illustrate the physical location of intermediary devices and cable installation.



Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network.



1.2 Common Types of Networks

Common Types of Networks

Networks of Many Sizes



Small Home



SOHO



Medium/Large



World Wide

- Small Home Networks – connect a few computers to each other and the Internet
- Small Office/Home Office – enables computer within a home or remote office to connect to a corporate network
- Medium to Large Networks – many locations with hundreds or thousands of interconnected computers
- World Wide Networks – connects hundreds of millions of computers world-wide – such as the internet

Common Types of Networks

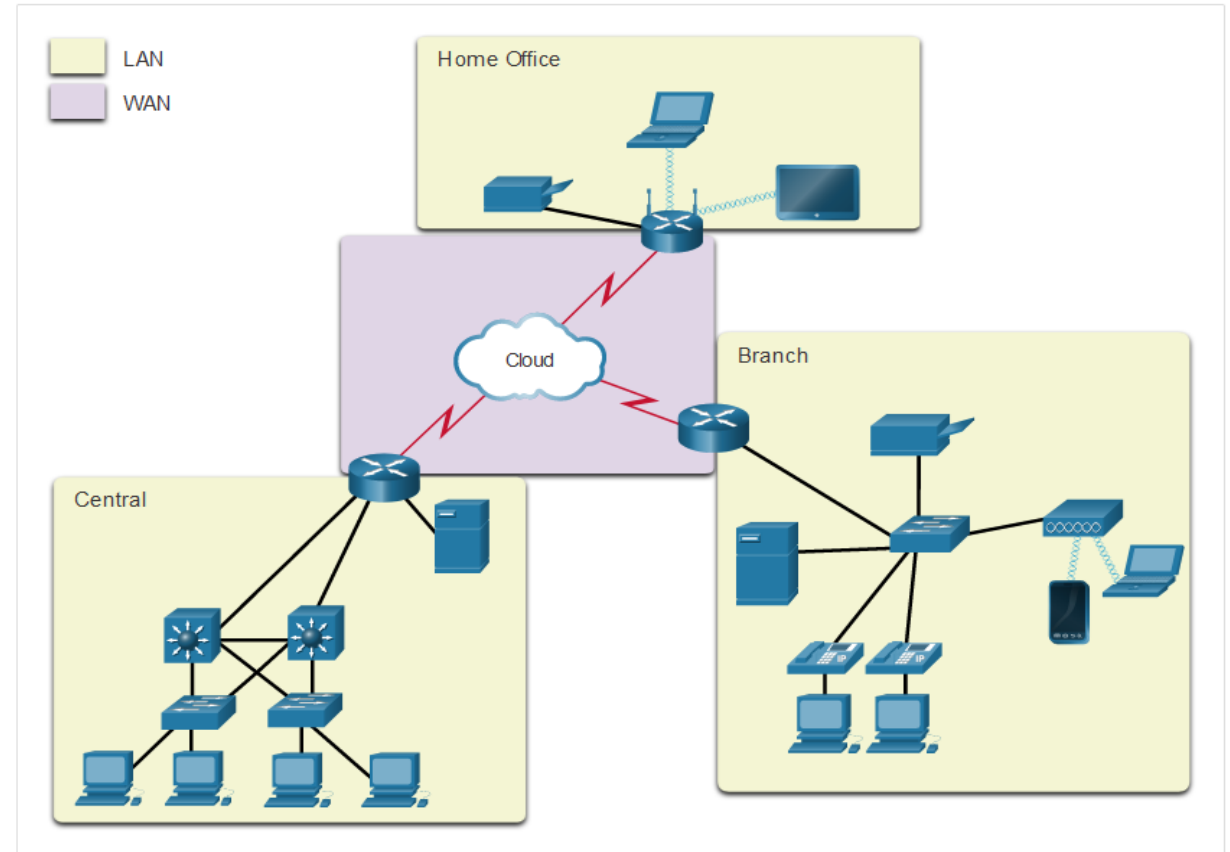
LANs and WANs

Network infrastructures vary greatly in terms of:

- Size of the area covered
- Number of users connected
- Number and types of services available
- Area of responsibility

Two most common types of networks:

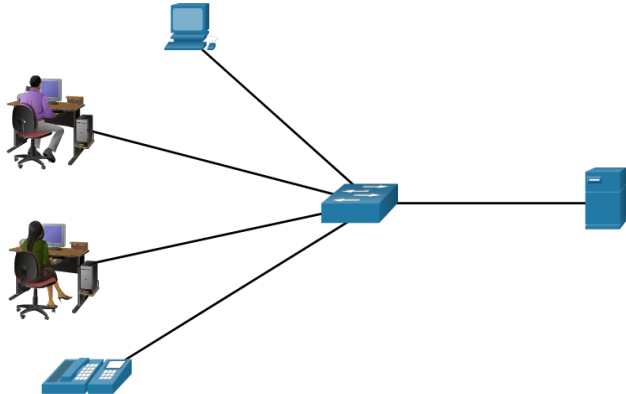
- Local Area Network (LAN)
- Wide Area Network (WAN).



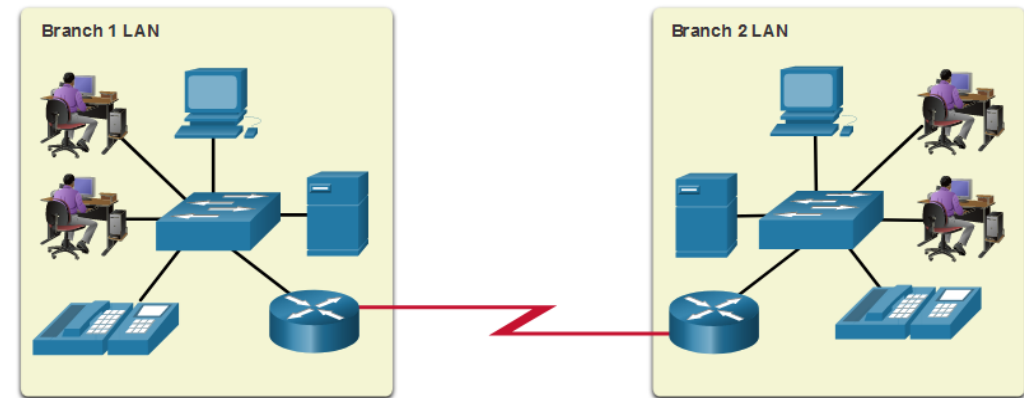
Common Types of Networks

LANs and WANs (cont.)

A LAN is a network infrastructure that spans a small geographical area.



A WAN is a network infrastructure that spans a wide geographical area.



LAN	WAN
Interconnect end devices in a limited area.	Interconnect LANs over wide geographical areas.
Administered by a single organization or individual.	Typically administered by one or more service providers.
Provide high-speed bandwidth to internal devices.	Typically provide slower speed links between LANs.

Common Types of Networks

The Internet

The internet is a worldwide collection of interconnected LANs and WANs.

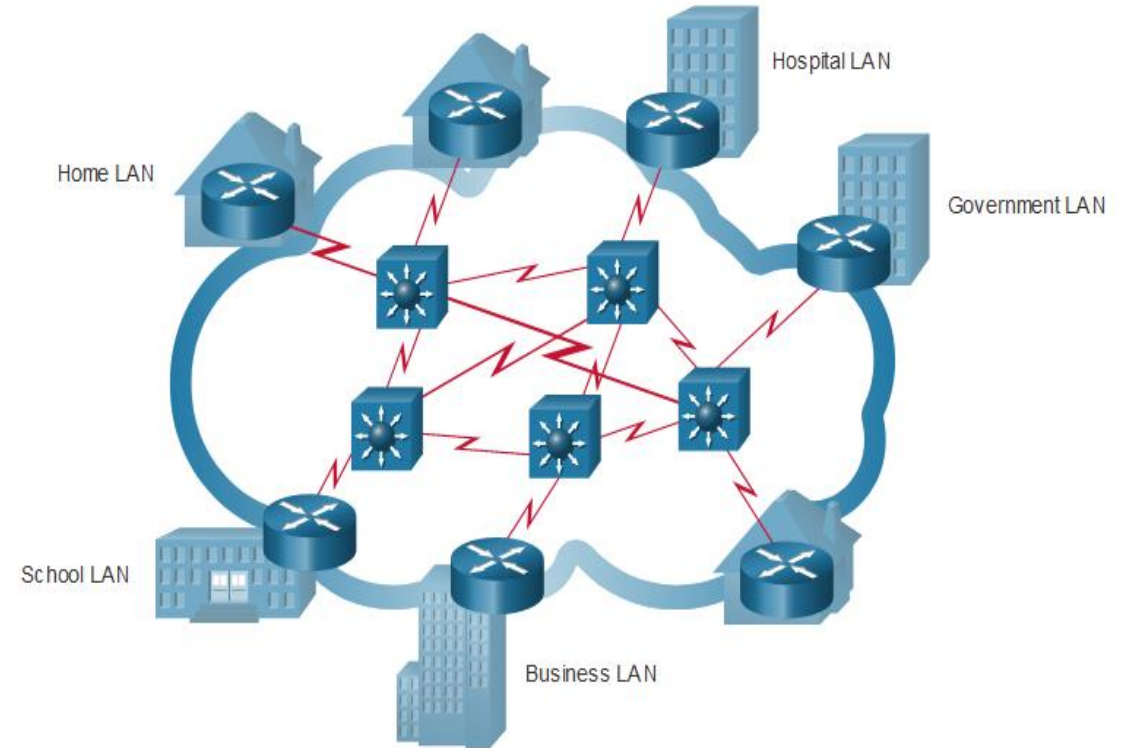
- LANs are connected to each other using WANs.
- WANs may use copper wires, fiber optic cables, and wireless transmissions.

The internet is not owned by any individual or group. The following groups were developed to help maintain structure on the internet:

IETF

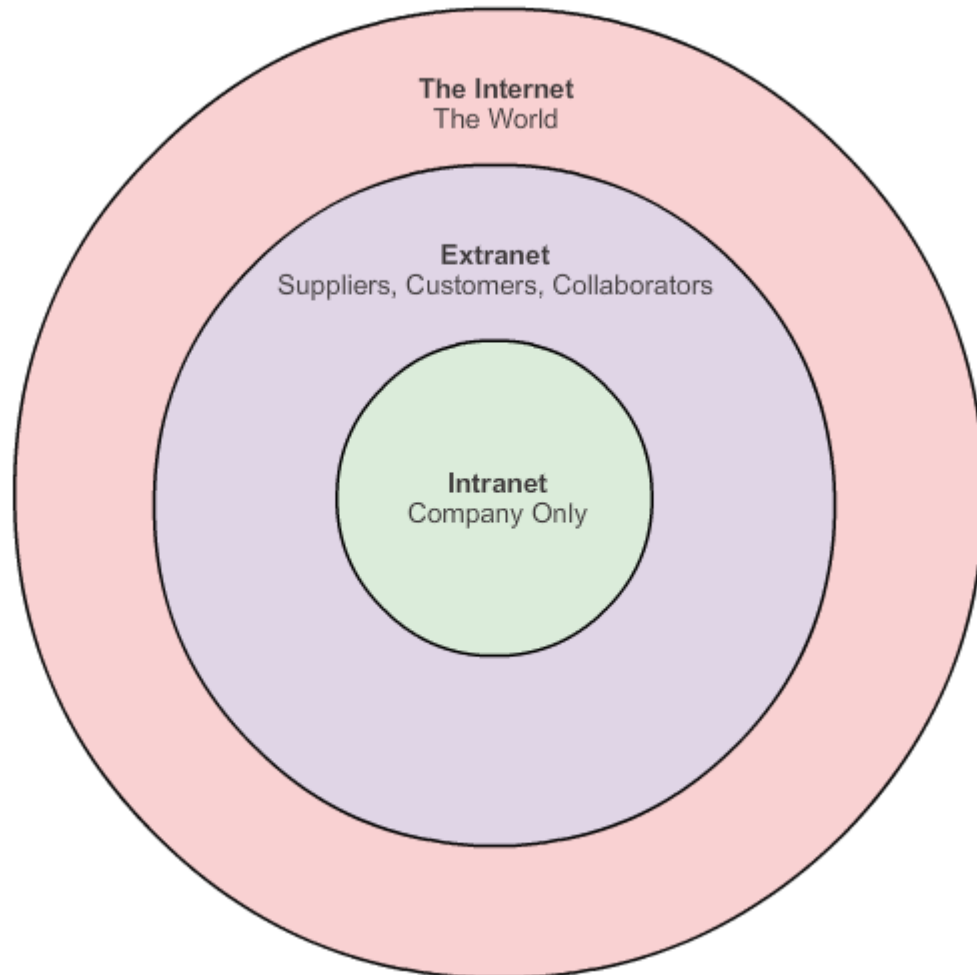
ICANN

IAB



Common Types of Networks

Intranets and Extranets



An **intranet** is a private collection of LANs and WANs internal to an organization that is meant to be accessible only to the organizations members or others with authorization.

An organization might use an **extranet** to provide secure access to their network for individuals who work for a different organization that need access to their data on their network.

1.3 Internet Connections

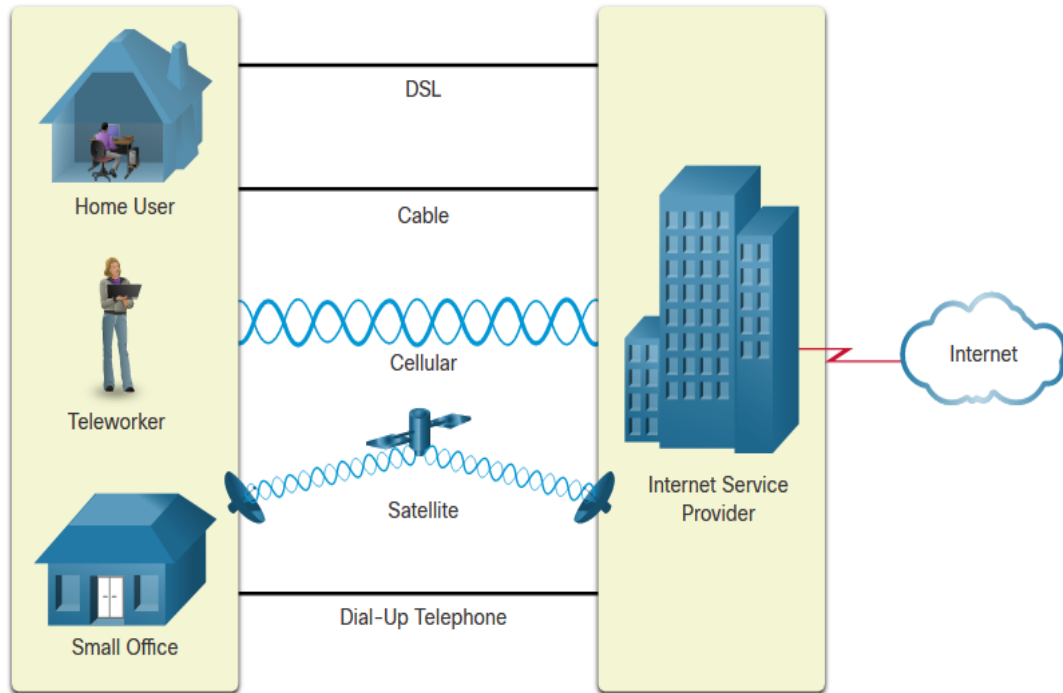
Internet Access Technologies



There are many ways to connect users and organizations to the internet:

- Popular services for home users and small offices include broadband cable, broadband digital subscriber line (DSL), wireless WANs, and mobile services.
- Organizations need faster connections to support IP phones, video conferencing and data center storage.
- Business-class interconnections are usually provided by service providers (SP) and may include: business DSL, leased lines, and Metro Ethernet.

Home and Small Office Internet Connections

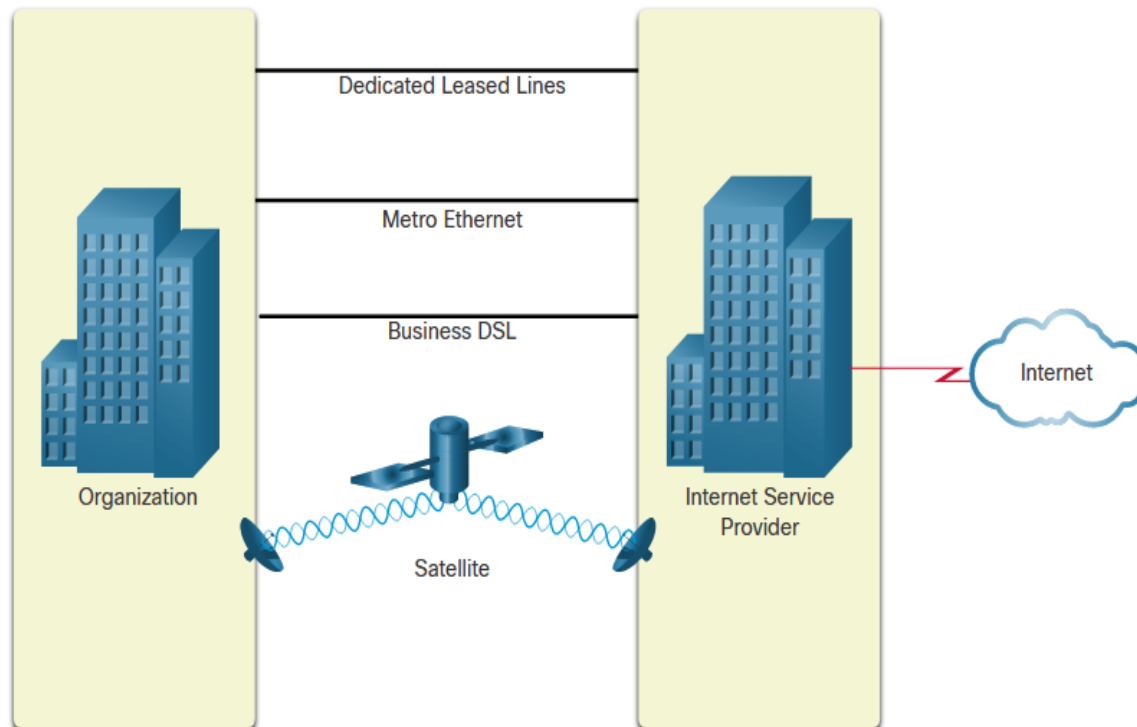


Connection	Description
Cable	high bandwidth, always on, internet offered by cable television service providers.
DSL	high bandwidth, always on, internet connection that runs over a telephone line.
Cellular	uses a cell phone network to connect to the internet.
Satellite	major benefit to rural areas without Internet Service Providers.
Dial-up telephone	an inexpensive, low bandwidth option using a modem.

Businesses Internet Connections

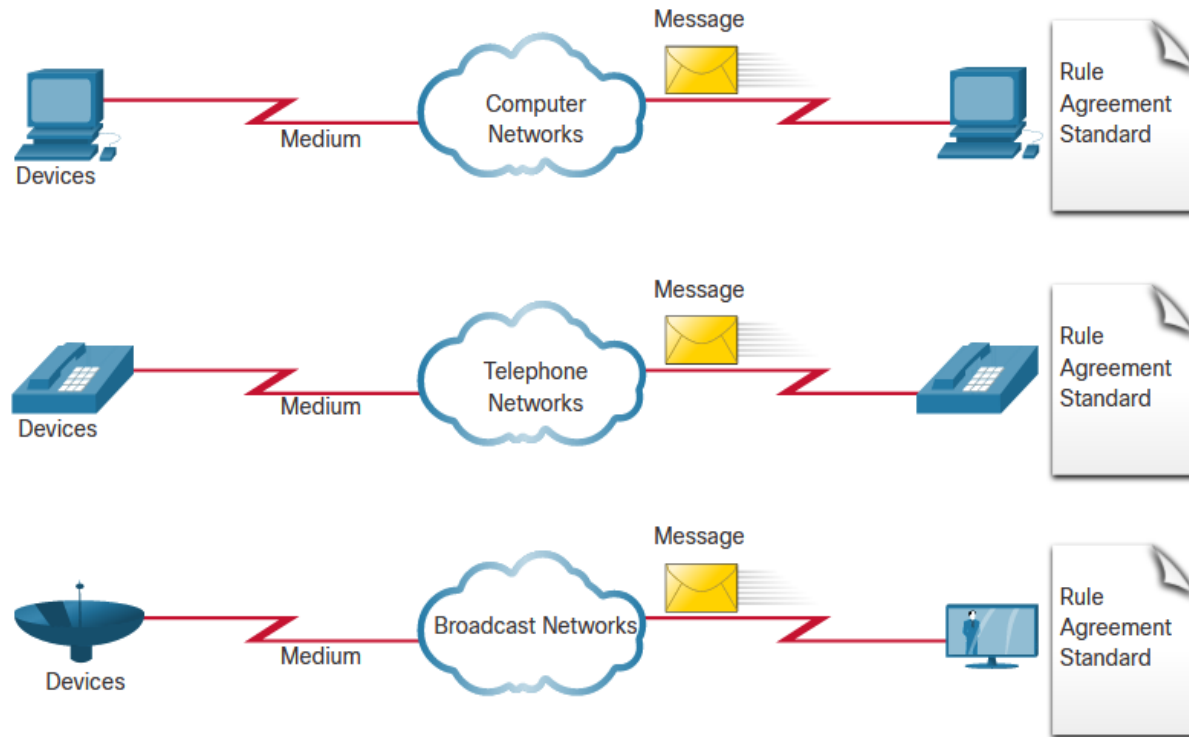
Corporate business connections may require:

- higher bandwidth
- dedicated connections
- managed services



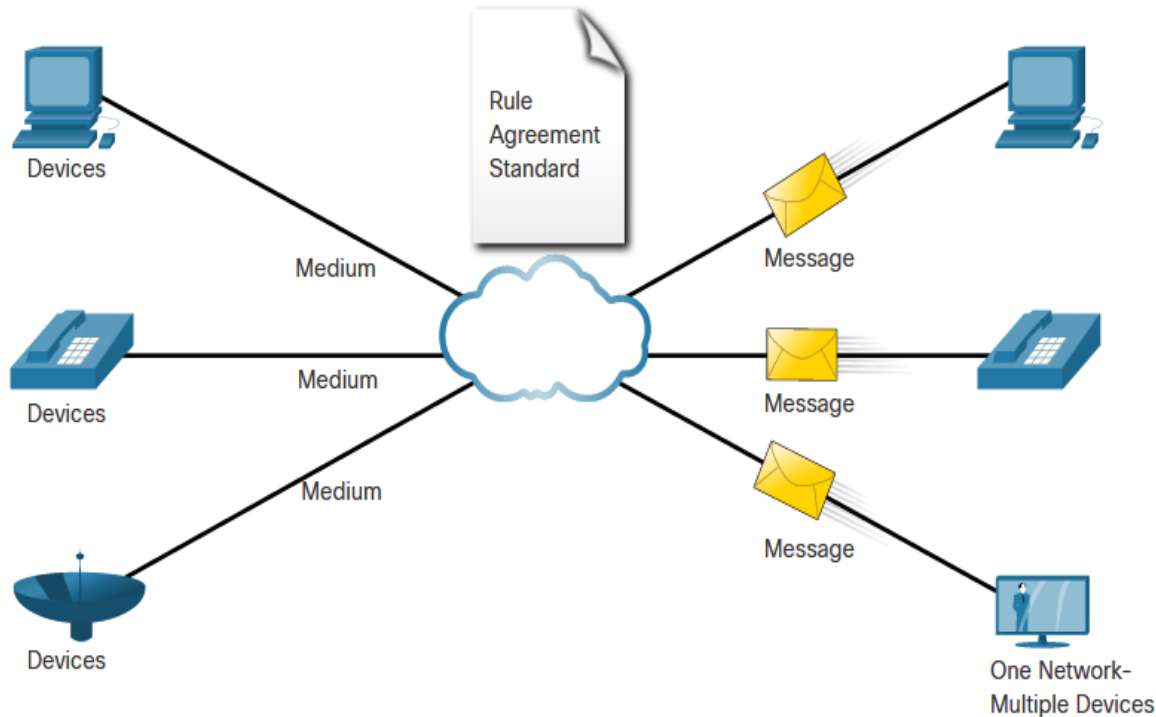
Type of Connection	Description
Dedicated Leased Line	These are reserved circuits within the service provider's network that connect distant offices with private voice and/or data networking.
Ethernet WAN	This extends LAN access technology into the WAN.
DSL	Business DSL is available in various formats including Symmetric Digital Subscriber Lines (SDSL).
Satellite	This can provide a connection when a wired solution is not available.

Traditional Separate Networks



- Before converged networks, an organization would have been separately cabled for telephone, video, and data.
- Each of these networks used different technologies to carry the signals.
- Each of these technologies would use a different set of rules and standards.

The Converging Network



- Converged data networks **carry multiple services** on one link including:
 - data
 - voice
 - video
- Converged networks can deliver data, voice, and video over the same network infrastructure.
- The network infrastructure **uses the same set of rules and standards**.

1.4 Reliable Networks

Reliable Network

Network Architecture



Network Architecture refers to the technologies that support the infrastructure that moves data across the network.

There are four basic characteristics that the underlying architectures need to address to meet user expectations:

- Fault Tolerance
- Scalability
- Quality of Service (QoS)
- Security

Reliable Network

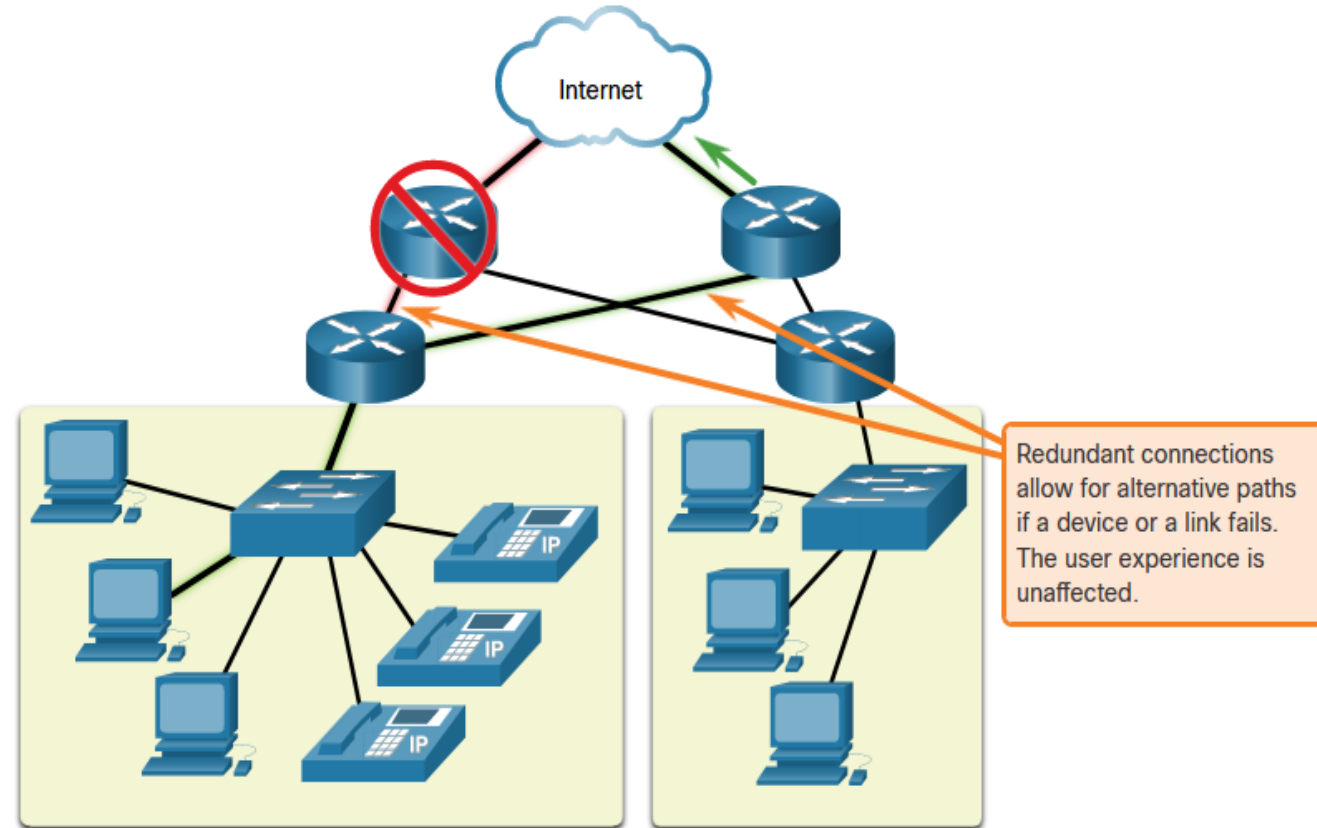
Fault Tolerance

A fault tolerant network limits the impact of a failure by limiting the number of affected devices. Multiple paths are required for fault tolerance.

Reliable networks provide redundancy by implementing a packet switched network:

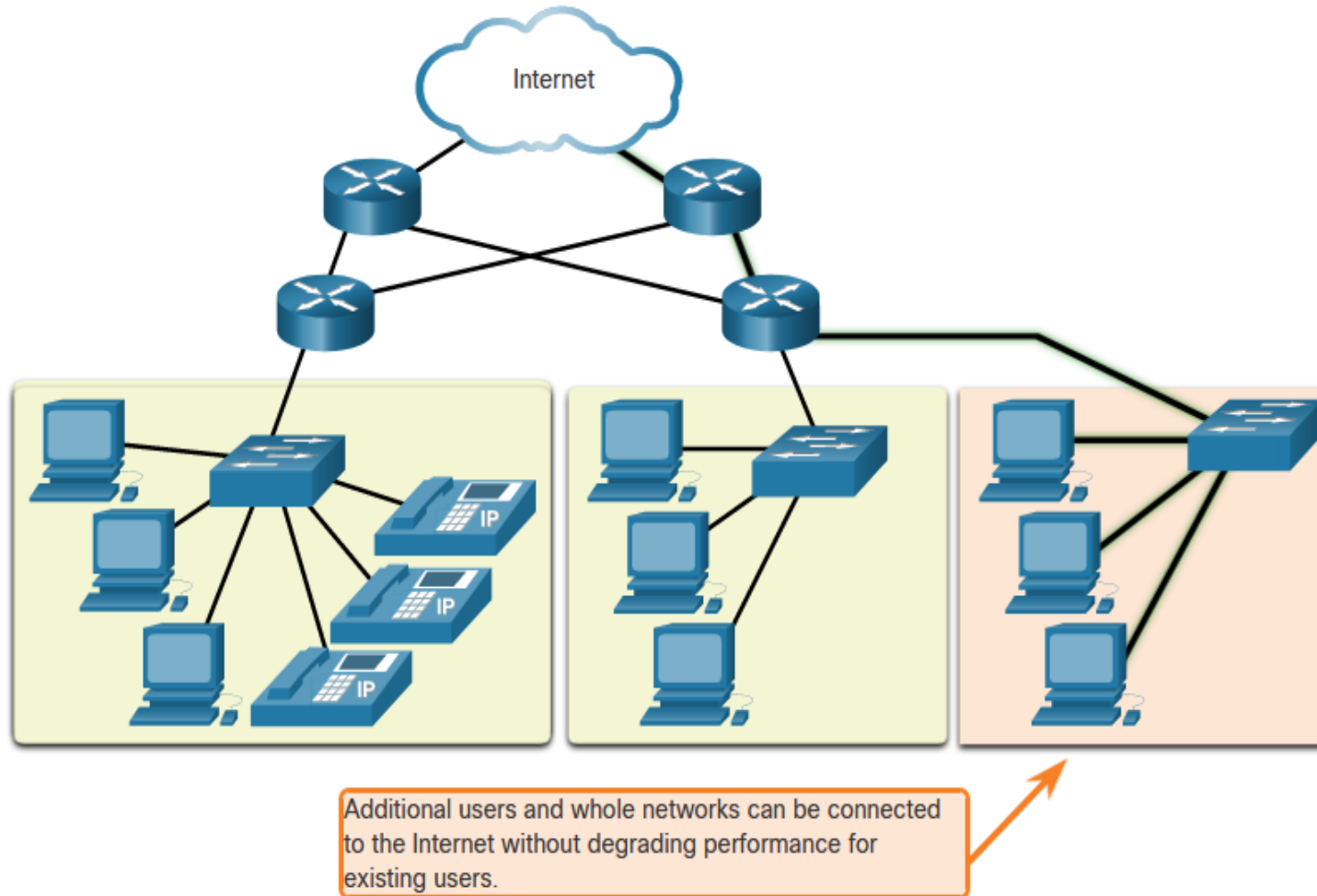
- Packet switching splits traffic into packets that are routed over a network.
- Each packet could theoretically take a different path to the destination.

This is not possible with circuit-switched networks which establish dedicated circuits.



Reliable Network

Scalability



A scalable network can expand quickly and easily to support new users and applications without impacting the performance of services to existing users.

Network designers follow accepted standards and protocols in order to make the networks scalable.

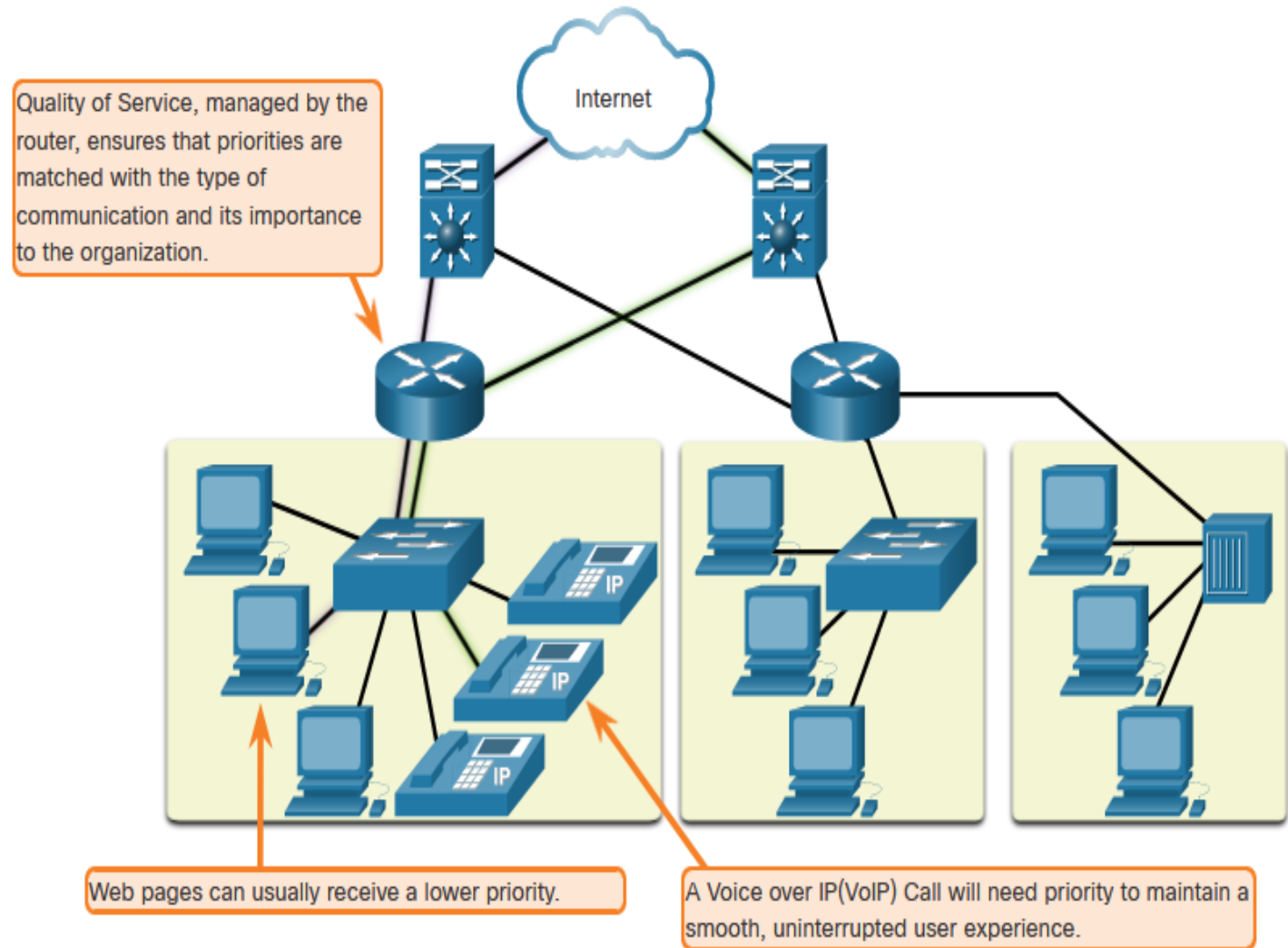
Reliable Network

Quality of Service

Voice and live video transmissions require higher expectations for those services being delivered.

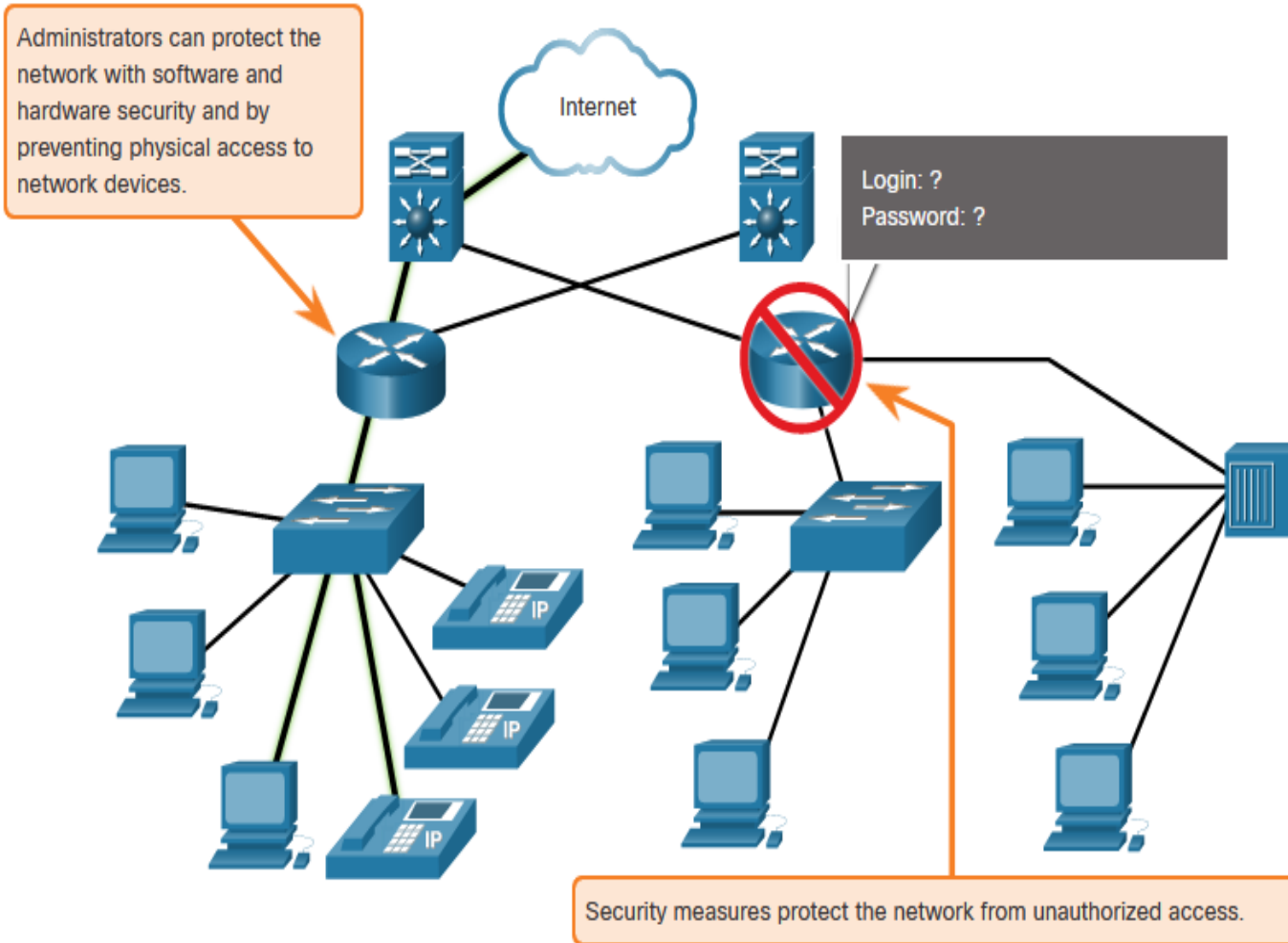
Have you ever watched a live video with constant breaks and pauses? This is caused when there is a higher demand for bandwidth than available – and QoS isn't configured.

- Quality of Service (QoS) is the primary mechanism used to ensure reliable delivery of content for all users.
- With a QoS policy in place, the router can more easily manage the flow of data and voice traffic.



Reliable Network

Network Security



There are two main types of network security that must be addressed:

- Network infrastructure security
 - Physical security of network devices
 - Preventing unauthorized access to the devices
- Information Security
 - Protection of the information or data transmitted over the network

Three goals of network security:

Confidentiality – only intended recipients can read the data

Integrity – assurance that the data has not be altered with during transmission

Availability – assurance of timely and reliable access to data for authorized users

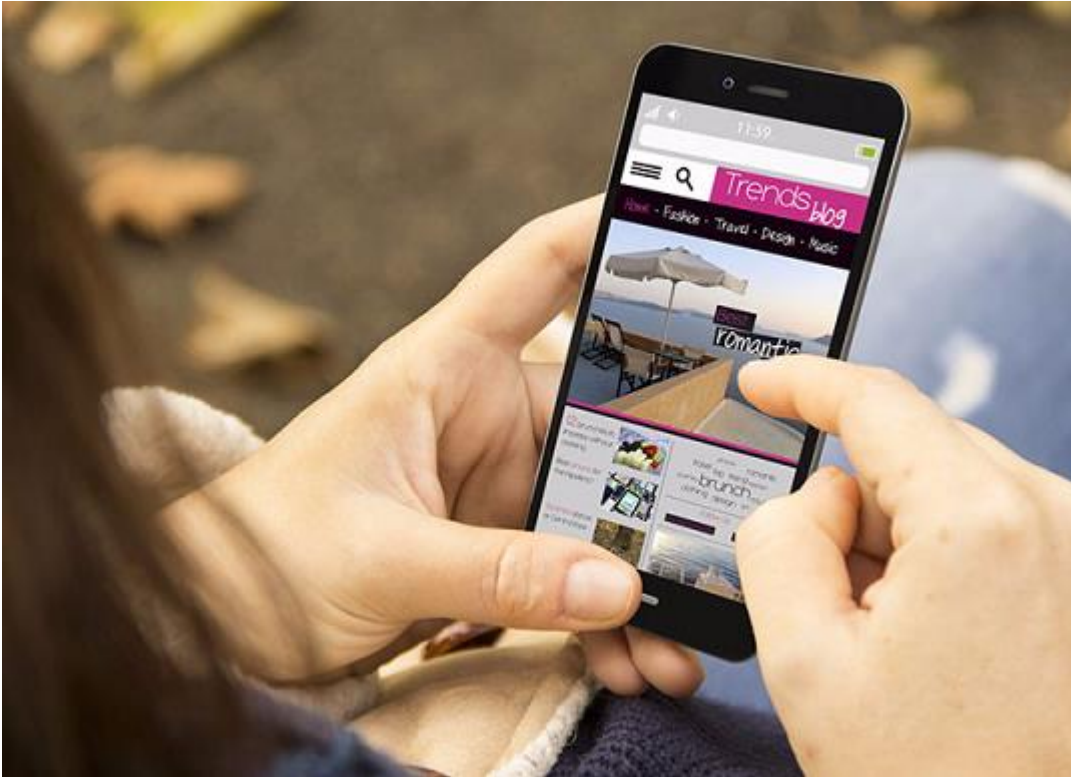
1.5 Networks Treads

New Trends



- The role of the network must adjust and continually transform in order to be able to keep up with new technologies and end user devices as they constantly come to the market.
- Several new networking trends that effect organizations and consumers:
 - Bring Your Own Device (BYOD)
 - Online collaboration
 - Video communications
 - Cloud computing

Bring Your Own Device



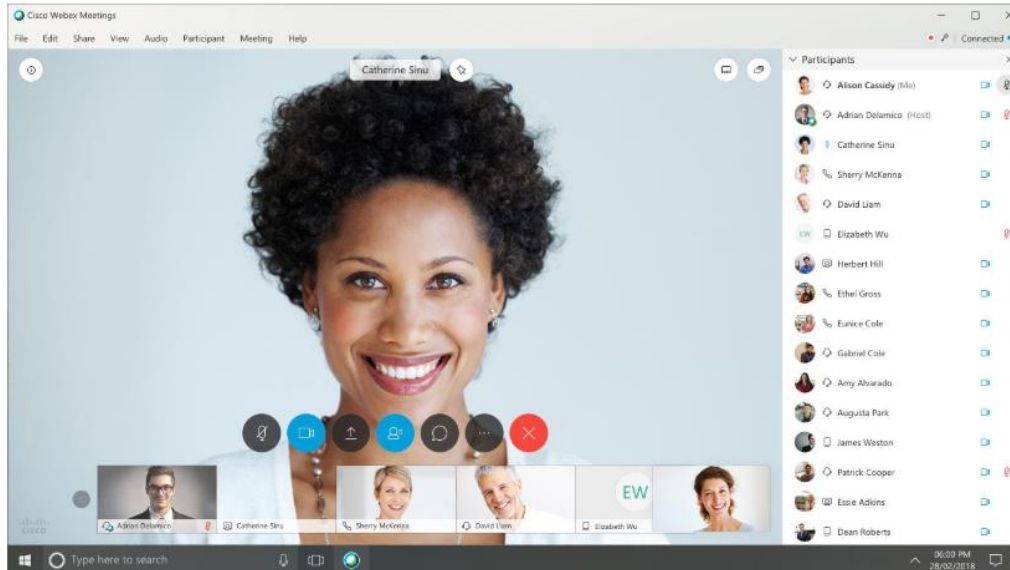
Bring Your Own Device (BYOD) allows users to use their own devices giving them more opportunities and greater flexibility.

BYOD allows end users to have the freedom to use personal tools to access information and communicate using their:

- Laptops
- Netbooks
- Tablets
- Smartphones
- E-readers

BYOD means any device, with any ownership, used anywhere.

Online Collaboration



- Collaborate and work with others over the network on joint projects.
- Collaboration tools including Cisco WebEx (shown in the figure) gives users a way to instantly connect and interact.
- Collaboration is a very high priority for businesses and in education.
- Cisco Webex Teams is a multifunctional collaboration tool.
 - send instant messages
 - post images
 - post videos and links

Video Communication

- Video calls are made to anyone, regardless of where they are located.
- Video conferencing is a powerful tool for communicating with others.
- Video is becoming a critical requirement for effective collaboration.
- Cisco TelePresence powers is one way of working where everyone, everywhere.



Cloud Computing

Cloud computing allows us to store personal files or backup our data on servers over the internet.

- Applications can also be accessed using the Cloud.
- Allows businesses to deliver to any device anywhere in the world.

Cloud computing is made possible by data centers.

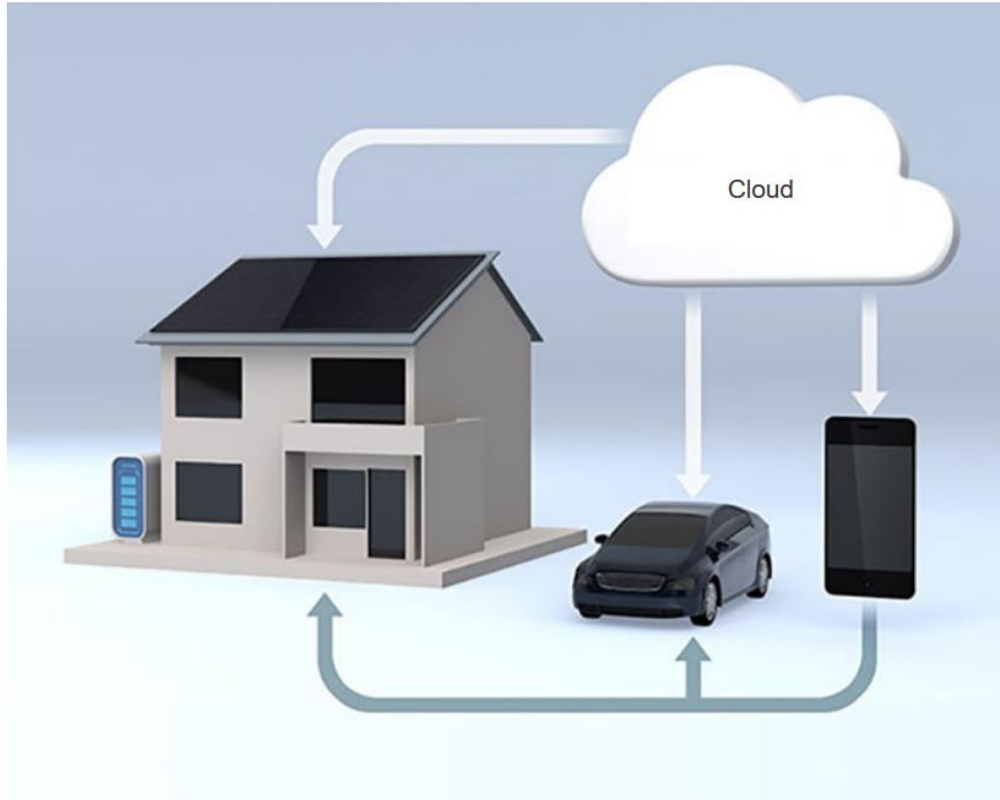
- Smaller companies that can't afford their own data centers, lease server and storage services from larger data center organizations in the Cloud.

Cloud Computing (Cont.)

Four types of Clouds:

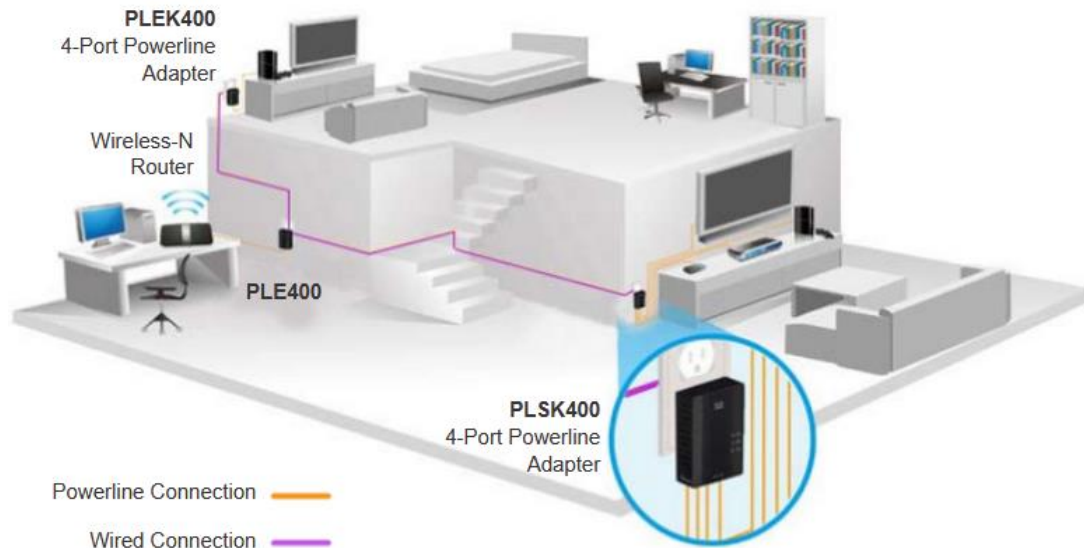
- Public Clouds
 - Available to the general public through a pay-per-use model or for free.
- Private Clouds
 - Intended for a specific organization or entity such as the government.
- Hybrid Clouds
 - Made up of two or more Cloud types – for example, part custom and part public.
 - Each part remains a distinctive object but both are connected using the same architecture.
- Custom Clouds
 - Built to meet the needs of a specific industry, such as healthcare or media.
 - Can be private or public.

Technology Trends in the Home



- Smart home technology is a growing trend that allows technology to be integrated into every-day appliances which allows them to interconnect with other devices.
- Ovens might know what time to cook a meal for you by communicating with your calendar on what time you are scheduled to be home.
- Smart home technology is currently being developed for all rooms within a house.

Powerline Networking



- Powerline networking can allow devices to connect to a LAN where data network cables or wireless communications are not a viable option.
- Using a standard powerline adapter, devices can connect to the LAN wherever there is an electrical outlet by sending data on certain frequencies.
- Powerline networking is especially useful when wireless access points cannot reach all the devices in the home.

Wireless Broadband



In addition to DSL and cable, wireless is another option used to connect homes and small businesses to the internet.

- More commonly found in rural environments, a Wireless Internet Service Provider (WISP) is an ISP that connects subscribers to designated access points or hotspots.
- Wireless broadband is another solution for the home and small businesses.
 - Uses the same cellular technology used by a smart phone.
 - An antenna is installed outside the house providing wireless or wired connectivity for devices in the home.

1.6 Issues of Network Management

Issues of Network Management

- Network administrator and engineer need to be aware of several common issues related to network management to effectively maintain and troubleshoot networks. Here are some of the key issues:

1. Network Performance

- **Bandwidth Congestion:** Overutilization of network bandwidth can lead to slow performance.
- **High Latency:** Delays in data transmission can affect the performance of real-time applications.
- **Packet Loss:** Data packets not reaching their destination can cause disruptions and data corruption.
- **Jitter:** Variability in packet arrival times can affect voice and video quality.

Issues of Network Management

2. Network Security

- **Unauthorized Access:** Protecting the network from unauthorized users.
- **Malware and Viruses:** Protecting the network from malicious software.
- **Data Breaches:** Preventing unauthorized access to sensitive data.
- **Denial of Service (DoS) Attacks:** Ensuring the network is resilient against DoS attacks.

3. Network Reliability

- **Hardware Failures:** Handling failures of network hardware components like routers, switches, and cables.
- **Redundancy and Failover:** Implementing redundant systems to ensure continuous operation.
- **Network Downtime:** Minimizing and managing planned and unplanned network outages.

Issues of Network Management

4. Configuration Management

- **Configuration Errors:** Ensuring network devices are correctly configured to avoid misconfigurations that can cause issues.
- **Documentation:** Keeping accurate records of network configurations, changes, and updates.
- **Change Management:** Carefully planning and documenting changes to the network to prevent disruptions.

5. Network Scalability

- **Capacity Planning:** Planning for future growth to ensure the network can handle increased load.
- **Upgrading Infrastructure:** Updating hardware and software to support new demands and technologies.
- **Load Balancing:** Distributing network traffic efficiently to avoid overloading any single component.

